Face-Position Detection using Landing Lens

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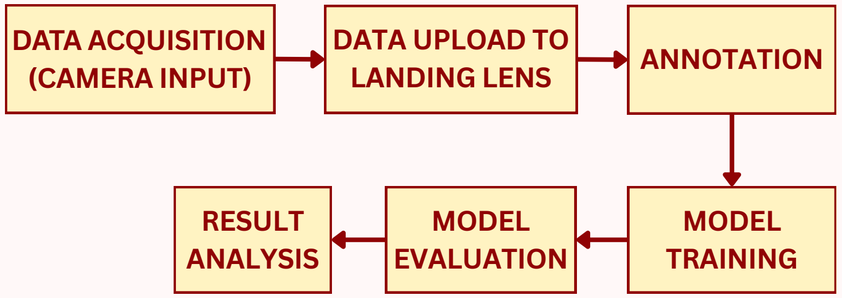
1. Abstract

This project develops a real-time, non-invasive facial position tracking system using precise head pose estimation to detect left, right, and front positions. The framework supports applications such as continuous patient monitoring, sleep pattern analysis, and ergonomic assessments, enhancing safety, operational efficiency, and healthcare outcomes.

2. Introduction

Facial position tracking provides non-intrusive, real-time monitoring of head orientation and posture using computer vision. It enhances healthcare, assisted living, and ergonomic applications while enabling advanced use cases such as sleep analysis and intelligent human–machine interaction, promoting safety, comfort, and operational efficiency.

3. Methodology



4. Input / Output

Test Image	Actual	Predicted	Result
	Front	Front	✓
	Right	Left	✗
	Right	Right	✓
	Front	Front	✓

5. Result

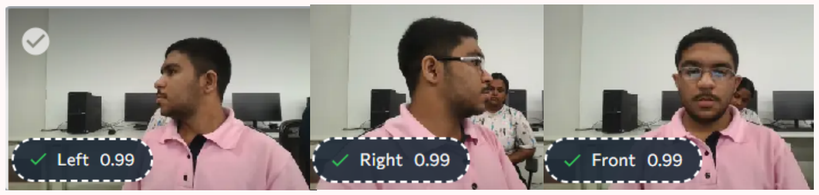
The trained model achieved high-precision facial position detection with less than 2% positional error. Visual tests confirmed accurate detection under partial occlusions and moderate head tilts, demonstrating suitability for real-time applications.

6. Conclusion

The system accurately detects and classifies head positions, enabling real-time monitoring for applications in security, attendance, and patient care. Its deployment supports efficient, non-intrusive monitoring and intelligent human–machine interaction.

7. Steps to use the Application

Step 1 : Take Sample Photos of face oriented in different positions- left, right and front.



Step 2: Scan the QR code.

Step 3: Upload the picture from gallery (or Take a live photo)

Step 4: Click "Use Photo" and get your result.

8. References

1. Zhang, Z., "A Flexible New Technique for Camera Calibration," IEEE Trans. Pattern Anal. Mach. Intell., 2000.
2. King, D. E., "Dlib-ml: A Machine Learning Toolkit," J. Machine Learning Research, 2009.
3. Viola, P., & Jones, M., "Rapid Object Detection using a Boosted Cascade of Simple Features," Proc. IEEE CVPR, 2001.
4. Landing AI - LandingLens Documentation- Official Website : support.landing.ai